

Exercise 1. Let G be a cyclic group of order n , with generator g . Prove that we construct a well-defined function from G to $\mathbb{Z}_n$ by setting $\varphi(g^k) = [k] \in \mathbb{Z}_n$. (This was a loose end left as an exercise in the Isomorphism notes.)

Collaborators: Max, Liantsoa, Patricia

Proof.

To show φ is well-defined, we need to show if $g^k = g^{k'}$ then $[k] = [k']$ in $\mathbb{Z}_n$.
Suppose $g^k = g^{k'}, g^k \in G$. Given that $|G| = |\langle g \rangle| = n$ so $g^n = e$
 $g^k g^{-k} = e$ because G is closed under inverse.

$$g^k g^{-k} = g^k (g^k)^{-1} = g^k (g^{k'})^{-1} = g^k g^{-k'} = g^{k-k'} = e$$

Notice $g^{k-k'}$ gives identity in the cyclic group G .

So, $k - k'$ is the multiple of n so that $g^{k-k'} = e$. (Proposition 7.7: Let $|g| = i$, $g^j = e$ iff $i \mid j$)

In other words, $n \mid k - k'$.

Since $k \equiv k'$ iff $n \mid k - k'$ by definition of $\equiv$ in $\mathbb{Z}_n$, $[k] = [k'] \pmod n$.

Therefore, φ is well-defined.

□

Exercise 2 Let G be an infinite cyclic group G with generator g . Let $m, n \in \mathbb{Z}$. Prove that $\langle g^m \rangle \cap \langle g^n \rangle$ is a cyclic subgroup of G , and find all of its generators.

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Proof. Since $\langle g^m \rangle \leq G$ and $\langle g^n \rangle \leq G$. The intersection of two subgroups, $\langle g^m \rangle \cap \langle g^n \rangle$, is also a subgroup of $G = \langle g \rangle$ (Proved in Exercise 5.7). Every subgroup of a cyclic group is cyclic (Proposition 7.6) and $\langle g^m \rangle \cap \langle g^n \rangle$ is a subgroup. Therefore, $\langle g^m \rangle \cap \langle g^n \rangle$ is a cyclic subgroup of G , denoted as H .

The generator in the group H must have the power that is multiple of both m and n . So all of its generator would be $g^{lcm(m,n)}$ and $g^{-lcm(m,n)}$.

□

Exercise 3 Prove that if $G_1 \cong H_1$ and $G_2 \cong H_2$, then $G_1 \times G_2 \cong H_1 \times H_2$.

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Proof. Given $G_1 \cong H_1$ and $G_2 \cong H_2$, S_0 , for $g_1, g'_1 \in G_1$ $g_2, g'_2 \in G_2$ we have $\varphi_1(g_1 g'_1) = \varphi_1(g_1) \varphi_1(g'_1)$ and $\varphi_2(g_2 g'_2) = \varphi_2(g_2) \varphi_2(g'_2)$

To prove $G_1 \times G_2 \cong H_1 \times H_2$, need to show that there exists a bijection f that respects binary operation.

$$f : G_1 \times G_2 \rightarrow H_1 \times H_2, \text{ where } f((g_1, g_2)) = (\varphi_1(g_1), \varphi_2(g_2))$$

one to one:

Suppose $g_1, g'_1 \in G_1$ and $g_2, g'_2 \in G_2$, $f[(g_1, g_2)] = f[(g'_1, g'_2)]$

then $(\varphi_1(g_1), \varphi_2(g_2)) = (\varphi_1(g'_1), \varphi_2(g'_2))$

Need to show $(g_1, g_2) = (g'_1, g'_2)$.

Since φ_1 and φ_2 are bijection and $\varphi_1(g_1) = \varphi_1(g'_1)$, $\varphi_2(g_2) = \varphi_2(g'_2)$; $g_1 = g'_1$ $g_2 = g'_2$ so $(g_1, g_2) = (g'_1, g'_2)$. It is one-to-one.

onto:

For any $(h_1, h_2) \in H_1 \times H_2$ and $h_1 \in H_1$, $h_2 \in H_2$, there exists $g_1 \in G_1$ s.t. $\varphi_1(g_1) = h_1$, and $g_2 \in G_2$ s.t. $\varphi_2(g_2) = h_2$ because φ_1, φ_2 are bijection.

$$f[(g_1, g_2)] = (\varphi_1(g_1), \varphi_2(g_2)) = (h_1, h_2)$$

So $f[(g_1, g_2)] = (h_1, h_2)$

$\therefore$ Bijection.

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Carry every properties of the group operation.

Let $(g_1, g_2), (g'_1, g'_2) \in G_1 \times G_2$ want to show: if $f((g_1, g_2)(g'_1, g'_2))$

then $f(g_1, g_2) f(g'_1, g'_2)$

$$f((g_1, g_2)(g'_1, g'_2)) = f((g_1 g'_1, g_2 g'_2))$$

$$= (\varphi_1(g_1 g'_1), \varphi_2(g_2 g'_2))$$

$$= (\varphi_1(g_1) \varphi_1(g'_1), \varphi_2(g_2) \varphi_2(g'_2))$$

$$= (\varphi_1(g_1), \varphi_2(g_2)) (\varphi_1(g'_1), \varphi_2(g'_2))$$

$$= f(g_1, g_2) f(g'_1, g'_2)$$

So, f respects the binary operation. Therefore, $G_1 \times G_2 \cong H_1 \times H_2$. □

Exercise 4 We have studied at least five groups of order 8 this semester: $D_4, \mathbb{Z}_8, Q_8, \mathbb{Z}_4 \times \mathbb{Z}_2$, and $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$. Prove that none of these groups is isomorphic to any of the others.

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Proof.

D_4 and Q_8 are nonabelian.

$\mathbb{Z}_8$, $\mathbb{Z}_4 \times \mathbb{Z}_2$, and $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ are abelian (component wise operation is addition, addition is commutative).

So, by Theorem 8.8, D_4 and Q_8 are not isomorphic to any other groups.

then focus on D_4 and Q_8 :

- D_4 has 4 rotation and 4 reflection. The reflections have order of 2.

-However Q_8 only has 2 element having order of 2, which are 1 and -1.

-so the orders of elements in each group do not match. D_4 is not isomorphic to Q_8

,and any other groups. Clean D_4 and Q_8 from the list.

Next, check $\mathbb{Z}_8$, $\mathbb{Z}_4 \times \mathbb{Z}_2$, and $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$:

$\mathbb{Z}_8$ is a cyclic group because it can be generated by a single element 1. But $\mathbb{Z}_4 \times \mathbb{Z}_2$, and $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ can not generated by a single element. For example, $(1, 0)$ cannot generate $(0, 1)$ and $(1, 0, 1)$ cannot generate $(1, 1, 1)$. So $\mathbb{Z}_8$ is not isomorphic to $\mathbb{Z}_4 \times \mathbb{Z}_2$, and $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$

Therefore, $\mathbb{Z}_8$ is kicked out.

Finally,

$$\mathbb{Z}_4 \times \mathbb{Z}_2 = \{(0, 0), (0, 1), (1, 0), (1, 1), (2, 0), (2, 1), (3, 0), (3, 1)\}$$

$$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 = \{(0, 0, 0), (0, 0, 1), (0, 1, 0), (0, 1, 1), (1, 0, 0), (1, 0, 1), (1, 1, 0), (1, 1, 1)\}$$

Every element in the $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ has order of 2. But $\mathbb{Z}_4 \times \mathbb{Z}_2$ has elements of order 4, for instance $(1, 0)$. So the orders of elements do not match between these last 2 groups.

So none of these groups is isomorphic to any of the others.

□